

Fine-mapped single-cell eQTL evaluation of AlphaGenome fine-tuning

1 Overview

We evaluate whether sequence-derived mutation effects distinguish fine-mapped single-cell expression quantitative trait loci (eQTLs). A positive is a variant with posterior inclusion probability (PIP) greater than 0.75 and a negative is a variant with PIP less than 0.01. The evaluation uses human fine-mapping panels and a 1 Mb sequence window centered on the transcription start site (TSS) of the associated gene.

The primary comparison is the pretrained AlphaGenome trunk with a cell-type-matched semantic RNA head versus the two best available joint fine-tuned models, a LoRA model and a LoRA+LoCon model. The benchmark retains up to 1,000 variants in each PIP class per fine-mapping panel with a fixed random seed. The raw and retained counts are recorded with the benchmark output.

2 Cell-type matching

The 35 fine-mapping panels use the prefixes Ast, End, Ext, IN, MG, OD, and OPC, with numbered panels denoting subpanels. These map to the Zemke 2023 RNA head as follows. Ast maps to ASC, End maps to Endo, MG maps to MGC, OD maps to ODC, and OPC maps to OPC. Ext maps to the mean of the eight excitatory tracks L2/3 IT, L4/5 IT, L5/6 NP, L5 ET, L5 IT, L6 CT, L6 IT CAR3, and L6b. IN maps to the mean of LAMP5, PVALB, SNCG, SST, and VIP. This is the closest available cell-type match because the pretrained model does not provide native single-cell RNA tracks with the fine-mapping labels.

3 Mutation effects

For variant i , let $\mathbf{y}_{i,r} \in \mathbb{R}^{B \times C}$ and $\mathbf{y}_{i,a} \in \mathbb{R}^{B \times C}$ be the reference and alternate RNA predictions at the model’s 128 bp resolution, where B is the number of output bins and $C = 19$ is the number of RNA tracks. Let $\mathcal{C}_g$ be the track indices for the matched cell group. The TSS score is

$$s_i^{\text{TSS}} = \frac{1}{|\mathcal{C}_g|} \sum_{c \in \mathcal{C}_g} |y_{i,a,b_{\text{TSS}},c} - y_{i,r,b_{\text{TSS}},c}|, \quad (1)$$

where b_{TSS} is the 128 bp bin containing the TSS. The gene score is the corresponding mean absolute alternate-reference difference after averaging each track over all 128 bp bins overlapping the annotated gene span,

$$s_i^{\text{gene}} = \frac{1}{|\mathcal{C}_g|} \sum_{c \in \mathcal{C}_g} \left| \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} y_{i,a,b,c} - \frac{1}{|\mathcal{B}_i|} \sum_{b \in \mathcal{B}_i} y_{i,r,b,c} \right|, \quad (2)$$

where $\mathcal{B}_i$ is the set of bins overlapping the gene span. Both scores are nonnegative because fine-mapping identifies variant probability, not the direction of expression change. Performance is measured by AUROC and average precision, overall and within bins of $|\text{variant} - \text{TSS}|$.

4 Results

The completed benchmark summary is shown in Table 1.

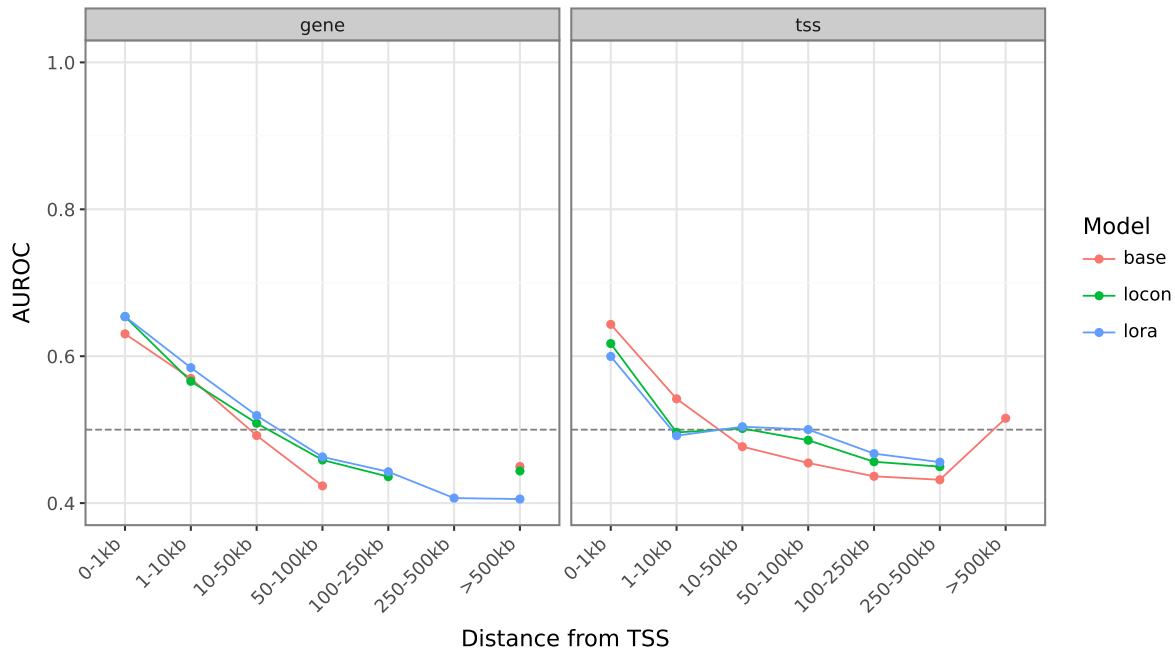


Figure 1: AUROC by distance from the TSS for TSS-bin and gene-span mutation effects. The dashed line is chance performance.

5 Implementation details

Fine-mapping tables are streamed from compressed PIP files. Variant alleles are checked against the reference genome before constructing the alternate one-hot sequence. The same reference and alternate sequences are evaluated for every model. The base comparator uses the pretrained trunk and the same 20-track RNA head, initialized by semantic bootstrap from compatible pretrained neural tracks; it is therefore a matched-head baseline rather than an uninitialized random head. Fine-tuned models retain their trained heads and adapters. All model prediction calls use the custom forward path so the fine-tuned heads, rather than the delegated pretrained variant scorer, are evaluated.

Table 1: Fine-mapped eQTL discrimination. Values are AUROC and average precision (AP); n_+ and n_- are the retained positive and negative counts. The table is generated from the benchmark output.

Model	Aggregation	Distance	AUROC	AP	n_+	n_-
base	TSS bin	50-100kb	0.455	0.442	3232	3432
base	TSS bin	500kb	0.516	0.086	126	1541
base	TSS bin	100-250kb	0.437	0.226	3558	10173
base	TSS bin	250-500kb	0.432	0.101	2253	16289
base	TSS bin	1-10kb	0.542	0.846	2947	616
base	TSS bin	10-50kb	0.477	0.666	5561	2640
base	TSS bin	0-1kb	0.643	0.981	1767	58
locon	TSS bin	50-100kb	0.486	0.470	3232	3432
locon	TSS bin	500kb	0.375	0.058	126	1541
locon	TSS bin	100-250kb	0.456	0.235	3558	10173
locon	TSS bin	250-500kb	0.450	0.105	2253	16289
locon	TSS bin	1-10kb	0.496	0.833	2947	616
locon	TSS bin	10-50kb	0.502	0.679	5561	2640
locon	TSS bin	0-1kb	0.617	0.981	1767	58
lora	TSS bin	50-100kb	0.500	0.483	3232	3432
lora	TSS bin	500kb	0.397	0.061	126	1541
lora	TSS bin	100-250kb	0.467	0.241	3558	10173
lora	TSS bin	250-500kb	0.456	0.107	2253	16289
lora	TSS bin	1-10kb	0.492	0.828	2947	616
lora	TSS bin	10-50kb	0.504	0.681	5561	2640
lora	TSS bin	0-1kb	0.600	0.980	1767	58
base	Gene span	50-100kb	0.423	0.431	3232	3432
base	Gene span	500kb	0.450	0.071	126	1541
base	Gene span	100-250kb	0.389	0.205	3558	10173
base	Gene span	250-500kb	0.387	0.093	2253	16289
base	Gene span	1-10kb	0.570	0.870	2947	616
base	Gene span	10-50kb	0.492	0.685	5561	2640
base	Gene span	0-1kb	0.630	0.980	1767	58
locon	Gene span	50-100kb	0.459	0.450	3232	3432
locon	Gene span	500kb	0.444	0.069	126	1541
locon	Gene span	100-250kb	0.436	0.222	3558	10173
locon	Gene span	250-500kb	0.396	0.095	2253	16289
locon	Gene span	1-10kb	0.566	0.867	2947	616
locon	Gene span	10-50kb	0.509	0.696	5561	2640
locon	Gene span	0-1kb	0.654	0.983	1767	58
lora	Gene span	50-100kb	0.463	0.450	3232	3432
lora	Gene span	500kb	0.406	0.067	126	1541
lora	Gene span	100-250kb	0.443	0.227	3558	10173
lora	Gene span	250-500kb	0.407	0.096	2253	16289
lora	Gene span	1-10kb	0.584	0.872	2947	616
lora	Gene span	10-50kb	0.519	0.699	5561	2640
lora	Gene span	0-1kb	0.654	0.982	1767	58
base	TSS bin	all	0.498	0.373	19444	34749
locon	TSS bin	all	0.509	0.383	19444	34749
lora	TSS bin	all	0.515	0.386	19444	34749
base	Gene span	all	0.497	0.394	19444	34749
locon	Gene span	all	0.523	0.413	19444	34749
lora	Gene span	all	0.533	0.416	19444	34749